When Silver Isn’t Gold: Quantifying and Auditing Group-Based Disagreement in Human and LLM Annotation

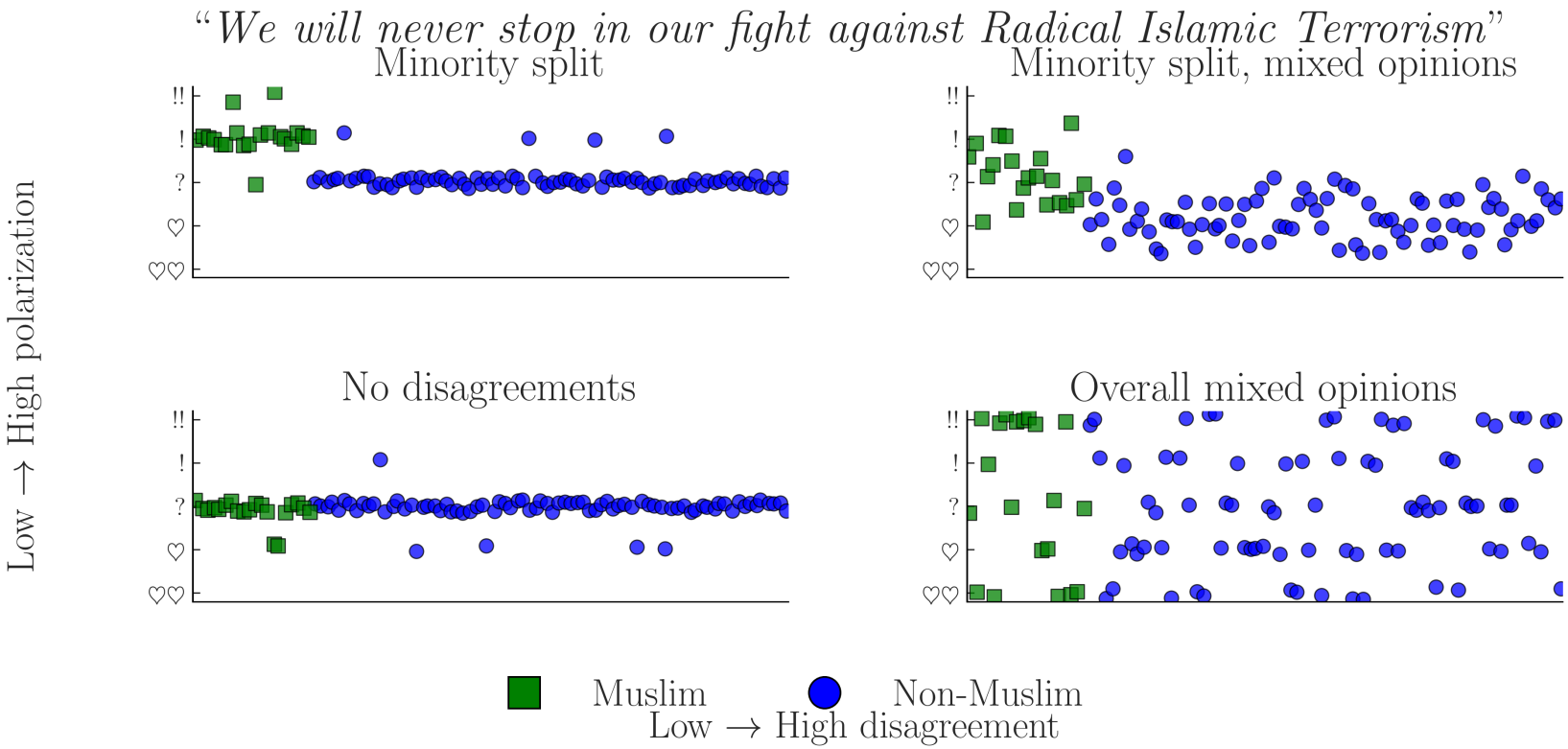
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Terminology

1. **Disagreement** represents any variance among annotators' ratings for the same item, regardless of its source.
2. **Systematic disagreement** is non-random and recurring disagreement, which in this work is assumed to be caused by differences in group identity (e.g., background, cultural context, ideology, sociodemographic characteristics).
3. **Polarization** is the underlying societal phenomenon describing genuinely conflicting, opposed viewpoints on the same item.

Goal


"We will never stop in our fight against Radical Islamic Terrorism"
Minority split
Minority split, mixed opinions
No disagreements
Overall mixed opinions
!!
!
?
♡
♡♡
Low → High polarization
Muslim
Non-Muslim
Low → High disagreement

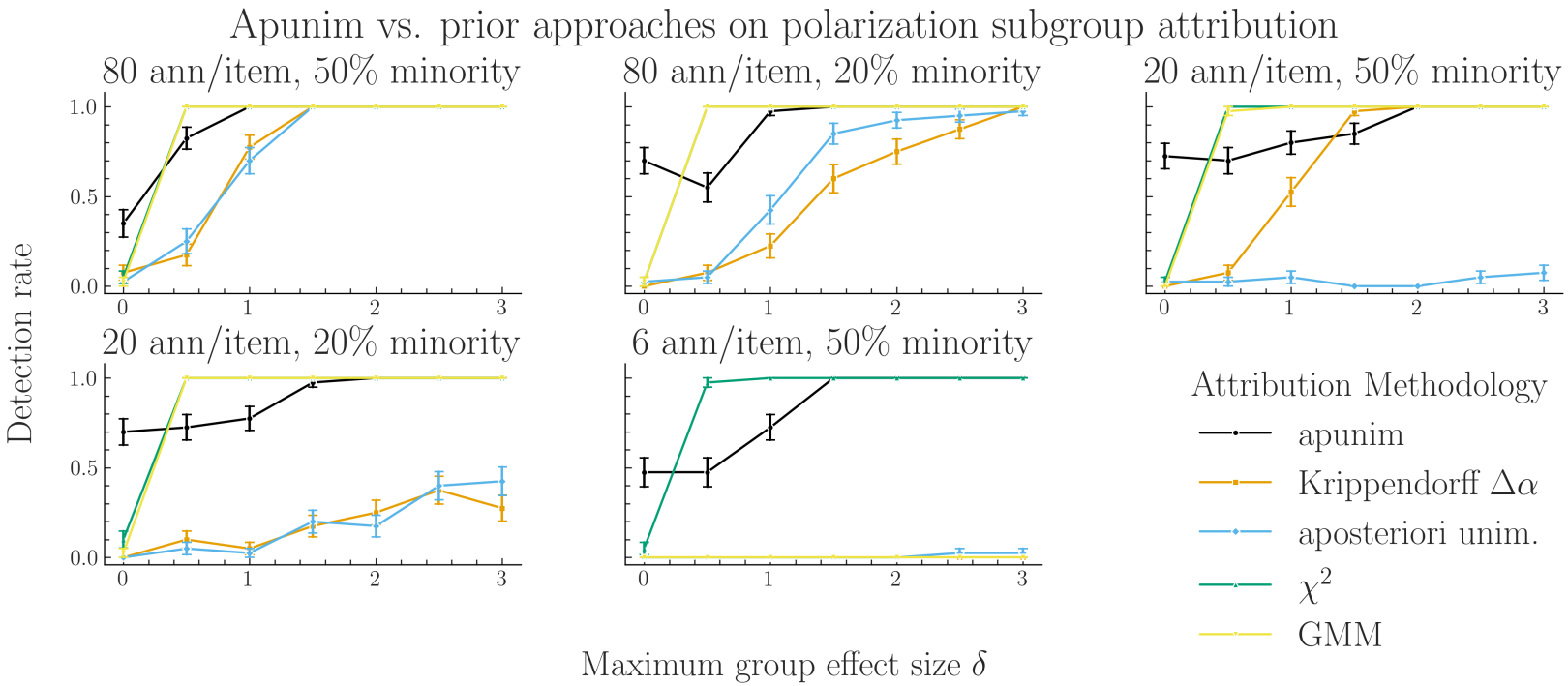
Lack of Datasets

Dataset	#Items	#Ann / Item	PCs	Text Source	Task	Guidelines
Kumar	1000	5	10	Twitter, Reddit, 4chan	Toxicity	Yes
Sap	585	5–6	3	Twitter	Hate Speech	No
DICES-350	350	123	4	LLM responses	Hate Speech	No
DICES-990	990	70–76	4	LLM responses	Hate Speech	No
POPQUORN	1500	7–9	5	Reddit	Hate Speech	Yes

Research Questions

1. How can we Quantitatively Attribute Systematic Disagreement to Annotator Groups?
2. Which Systematic Disagreement Patterns are Shared Between Human and LLM Annotators?

On Synthetic Data

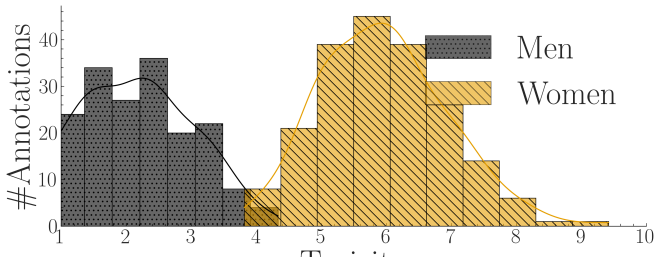


On Real-Life Data

	DICES-350				DICES-990				Sap et al. 2022			Kumar et al. 2021											POPQUORN				
Method	Age	Education	Gender	Race	Age	Education	Gender	Race	Age	Ethnicity	Gender	Age	Education	Ethnicity	Gender/Trans	Targeted	Parent	Politics	Religion	Tox.Seen/Problem	Sex. Orient.	Tech Impact	Age	Education	Gender	Occupation	Race
am	×	✓	×	✓	✓	✓	✓	✓	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	✓
$\Delta\alpha$	×	×	✓	✓	×	✓	✓	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
AU	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
χ^2	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

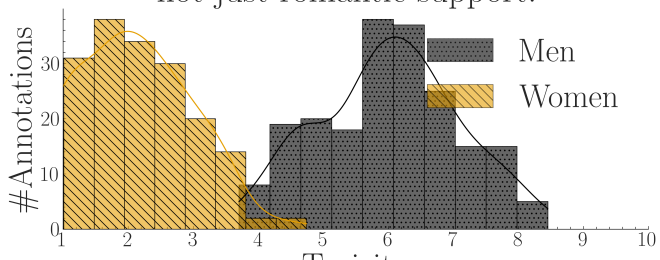
Why? Aggregation

"Most women do not drive well."



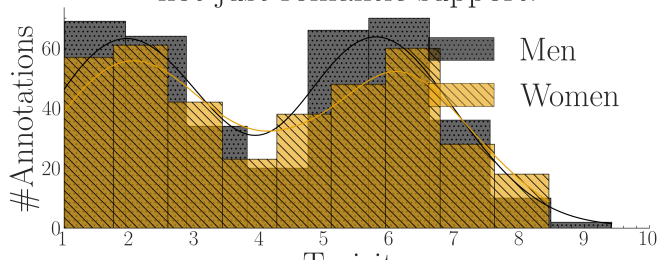
$nDFU_{Men} = 0.1944$
$nDFU_{Women} = 0.0000$
$nDFU_{All} = 1.0000$

"Most men need closer friendships, not just romantic support."



$nDFU_{Men} = 0.0526$
$nDFU_{Women} = 0.0000$
$nDFU_{All} = 1.0000$

"Most women do not drive well."
"Most men need closer friendships, not just romantic support."

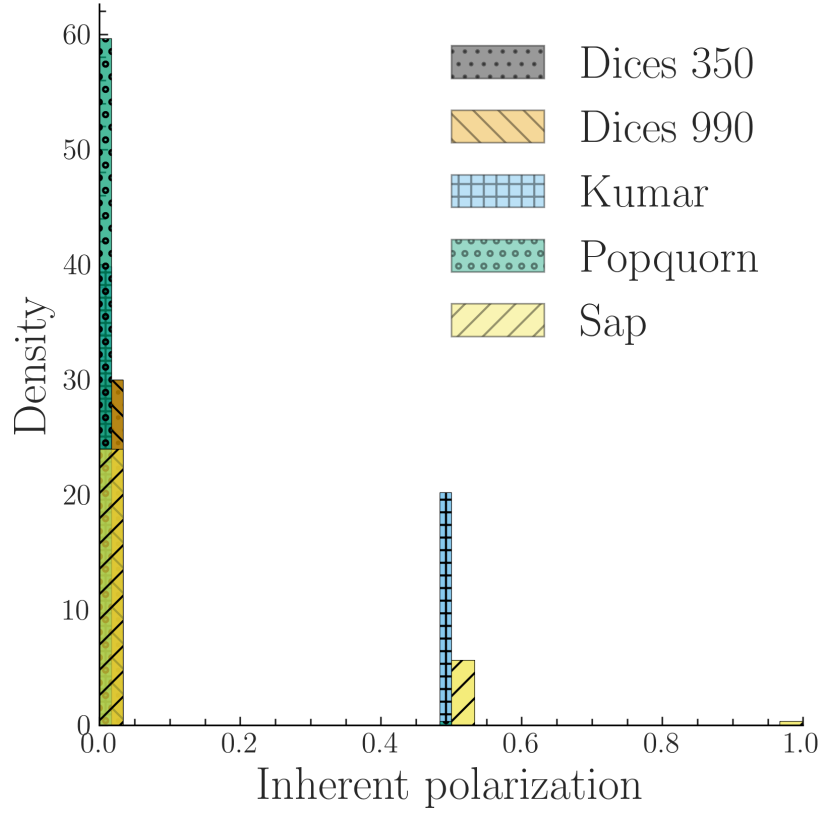


$nDFU_{Men} = 0.4286$
$nDFU_{Women} = 0.2459$
$nDFU_{All} = 1.0000$

Why? Unattributable Polarization

$$\mathrm{Pol}_{inh}(c) = \min_{S \in G_c} \{\mathrm{nDFU}(S)\}$$

$$\mathrm{Pol}^{MC}_{inh}(c) = \min_{j=1}^{k} \{\mathrm{nDFU}(S_j)\}$$



Quantifying Attribution to Annotator Groups

$$apunim(D, g) = \frac{\mathbb{E}_{c \in D}\Big[\mathrm{Pol}_{apr}(c)\Big] - \mathbb{E}_{c \in D}\Big[Pol_g(c)\Big]}{1 - \mathbb{E}_{c \in D}\Big[\mathrm{Pol}_{apr}(c)\Big]}$$

How Are Humans Polarized?

Group	Apunim	Support
DICES-990		
Ethnicity=AfricanAm	0.099	5148
Ethnicity=Latino	-0.063	5877
Ethnicity=Other	0.034	21669
Ethnicity=White	-0.065	10166
Gender=Man	0.036	28817
Gender=Woman	-0.027	29990
Age=GenZ	0.047	12214
Age=Millennial	-0.007	34433
Age=GenX+	-0.028	12699
Education=College	0.006	52727
Education=High school	-0.050	6619

Group	Apunim	Support
DICES-350		
Ethnicity=AfricanAm	-0.037	7627
Ethnicity=Asian	0.045	6838
Education=Other	-0.023	1841
Sap		
Gender=Man	0.0907	1439
Gender=Woman	-0.2442	877
POPQUORN		
Gender=Man	0.0081	2720
Race=Black	-0.1572	222
Education=College	0.0496	2388
Education=Graduate	-0.0808	468

How Are LLMs Polarized? Unimodality

	Sap et al. 2022		Kumar et al. 2021		POPQUORN		All	
	n	Mean (SE)	n	Mean (SE)	n	Mean (SE)	n	Mean (SE)
Model								
Human	585	0.38 (0.01)	1000	0.42 (0.01)	1500	0.31 (0.01)	3085	0.36 (0.00)
olmo32b	585	0.24 (0.01)	1000	0.51 (0.01)	1500	0.50 (0.01)	3085	0.45 (0.01)
qwen32b	585	0.18 (0.01)	1000	0.19 (0.01)	1499	0.20 (0.01)	3084	0.19 (0.00)
mistral7b	585	0.21 (0.01)	—	—	1500	0.40 (0.01)	2085	0.34 (0.01)
mistral24b	585	0.25 (0.01)	—	—	1500	0.23 (0.01)	2085	0.24 (0.01)
gemma7b	585	0.19 (0.01)	—	—	1500	0.12 (0.01)	2085	0.14 (0.01)
gemma31b	585	0.04 (0.01)	—	—	1500	0.05 (0.00)	2085	0.05 (0.00)

How Are LLMs Polarized? Unattributable

Model	Dataset	n	Stereotype	Persona	Single	Direct
olmo32b	All	3085	-0.09	-0.02	0.01	-0.15
	Sap et al. 2022	585	0.09	0.06	0.22	0.04
	Kumar et al. 2021	1000	0.14	-0.13	-0.29	-0.31
	POPQUORN	1500	-0.50	-0.00	0.10	-0.19
qwen32b	All	3084	0.25	0.22	0.22	0.19
	Sap et al. 2022	585	0.32	0.13	-0.09	0.09
	Kumar et al. 2021	1000	0.17	0.21	0.51	0.31
	POPQUORN	1499	0.26	0.30	0.24	0.17
mistral7b	All	2085	0.16	0.06	0.21	0.10
	Sap et al. 2022	585	0.21	0.10	0.08	0.57
	POPQUORN	1500	0.12	0.02	0.33	-0.36
mistral24b	All	2085	0.60	0.12	-0.53	0.31
	Sap et al. 2022	585	0.64	0.04	-0.76	0.27
	POPQUORN	1500	0.55	0.20	-0.30	0.35
gemma7b	All	2085	0.26	0.51	0.78	0.39
	Sap et al. 2022	585	0.18	0.48	0.57	0.15
	POPQUORN	1500	0.34	0.54	0.99	0.63
gemma31b	All	2085	0.42	0.31	0.00	0.63
	Sap et al. 2022	585	0.36	0.21	0.06	1.03
	POPQUORN	1500	0.47	0.40	-0.06	0.23

How Are LLMs Polarized? Group Attribution

		olmo32b					qwen32b					mistral7b					mistral24b					gemma7b					gemma31b				
Dataset	PC																													
Sap et al. 2022	Age	×	×	×	✓	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	✓	✓	×	✓	×	×	×	×	×	×
	Ethnicity	✓	✓	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×
	Gender	×	×	×	×	×	×	✓	×	×	±	×	✓	✓	×	×	±	✓	✓	✓	±	×	×	±	×	±	×	×	×	×	✓
POPQUORN	Age	×	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	✓
	Education	✓	✓	±	✓	✓	±	±	±	✓	±	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	×	✓	✓	✓	×	±	±	×	×
	Gender	✓	✓	✓	✓	×	✓	×	×	×	×	×	✓	×	✓	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×	×
	Occupation	×	×	✓	✓	✓	✓	✓	✓	×	×	±	±	±	✓	✓	✓	±	✓	✓	✓	✓	±	±	✓	×	×	×	✓	×	×
	Race	✓	×	✓	✓	✓	±	±	±	✓	✓	✓	±	±	✓	±	±	±	✓	✓	±	±	±	✓	±	±	±	±	±	✓	✓